Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *MASSIVE* | ***Context:*** *Ethiopian Intercity Travelers — Amharic-Language Booking Bot*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotators were vendor-pool MTurk workers selected via a listening-based fluency test [Q51], a three-judgment quality test [Q52], and a translator quiz with 'customized local-language examples' [Q53]; workers self-certified as 'native speaker or … fluent in the required language' [Q122]. Annotator demographics — whether Amharic translators are urban Addis residents, Amhara-region speakers, or L2 speakers — are not disclosed (regional context, amharic_dialect_variation.massive_annotator_dialect_representation: NOT FOUND). The paper acknowledges 'the lack of labeled data … that is realistic for the task and that is natural for each given language' [Q6] as the central problem the dataset attempts to address — but the translation-from-English pipeline means Amharic instances reflect English-language pragmatics translated into Amharic [Q2, Q4]. The user has confirmed code-switching with Oromo (Q3 elicitation) will occur in real utterances, and gaps 3 and 4 confirm no Oromo–Amharic or Tigrinya–Amharic code-switching booking-domain corpus exists. 'Some low-quality utterances' are acknowledged [Q104] and removable only at the cost of further reducing the already small per-language sample [Q103, Q105]. OC was rated HIGH priority and the violation is severe.

ID	Evidence
Q2	"MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera." (p.1)
Q3	"With the English seed data included, there are 587k train utterances, 104k dev utterances, 152k test utterances, and 153k utterances currently held out for the MMNLU-22 competition, which will be released after the competition." (p.1)
Q4	"MASSIVE was created by localizing the SLURP NLU dataset (created only in English) in a parallel manner." (p.1)
Q6	"one difficulty in creating massively multilingual NLU models is the lack of labeled data for training and evaluation, particularly data that is realistic for the task and that is natural for each given language." (p.1)
Q51	"We offered two mechanisms to vendors for evaluating workers to be selected for each language. The first, which was used to select workers for the translation task, was an Amazon MTurk-hosted fluency test where workers listen to questions and statements in the relevant language and were evaluated using a multiple-choice questionnaire." (p.6)
Q52	"The second, which was used to select workers for the judgment task, was a test with a set of three judgments that the vendor could use to assess if workers were able to detect issues in the translated utterances." (p.6)
Q53	"In order to further improve worker selection quality, we created a translator quiz using the Amazon MTurk instructions that were created for translation and judgment tasks, coupled with customized local-language examples." (p.6)
Q103	"Starting with the dataset, the per-language data quantities are relatively small at 19.5k total records and 11.5k records for training." (p.9)
Q104	"Second, there are some low-quality utterances, both in the seed data and in the translations." (p.9)
Q105	"For the most part, these are surfaced through the judgment scores we provide for each record, but if a user does filtering based on these judgments, then the data size decreases even further." (p.9)
Q122	"By clicking "SUBMIT", I also certify that I am a native speaker or am fluent in the required language" (p.17)

Strengths

- Multi-stage quality controls: fluency test [Q51], three-judgment QA [Q52], translator quiz [Q53], MT-detection alarms [Q66]
- Three independent judgment scores per utterance covering intent match, slot match, grammaticality, spelling, language identification [Q63] — included in metadata [Q64] for filtering

- Self-certified native/fluent speakers [Q122] and vendor monitoring for engagement quality [Q44, Q66]

ID	Evidence
Q44	“This vendor pool was reduced to three based on engagement and resource availability.” (p.5)
Q51	“We offered two mechanisms to vendors for evaluating workers to be selected for each language. The first, which was used to select workers for the translation task, was an Amazon MTurk-hosted fluency test where workers listen to questions and statements in the relevant language and were evaluated using a multiple-choice questionnaire.” (p.6)
Q52	“The second, which was used to select workers for the judgment task, was a test with a set of three judgments that the vendor could use to assess if workers were able to detect issues in the translated utterances.” (p.6)
Q53	“In order to further improve worker selection quality, we created a translator quiz using the Amazon MTurk instructions that were created for translation and judgment tasks, coupled with customized local-language examples.” (p.6)
Q63	“The output of the second workflow (the fully localized utterance) is judged by three workers for (1) whether the utterance matches the intent semantically, (2) whether the slots match their labels semantically, (3) grammaticality and naturalness, (4) spelling, and (5) language identification—English or mixed utterances are acceptable if that is natural for the language, but localizations without any tokens in the target language were not accepted.” (p.6)
Q64	“These judgments are also included in the metadata of the dataset.” (p.6)
Q66	“Workers were also monitored to see if their tasks were primarily machine translated. Such workers were removed from the pool and all of their work was resubmitted to be completed by the other workers.” (p.6)
Q122	“By clicking "SUBMIT", I also certify that I am a native speaker or am fluent in the required language” (p.17)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — labels reflect English-source intent/slot judgments translated into Amharic [Q2, Q4]; they do not reflect Ethiopian intercity-traveler stakeholder perspectives. Code-switching, L2-speaker phrasing, and operator-specific slot judgments are not represented (gaps 3, 4, 5).

OC-2: Substantial disagreement is likely between MASSIVE Amharic annotators (translators of English seed data) and the actual Ethiopian intercity-traveler population, particularly for code-switched and Ethiopian-calendar-bearing utterances (Q3 elicitation; gaps 3, 4, 5).

OC-3: Demographic information for Amharic translators is not disclosed in the paper — workers are described only as fluency-tested vendors [Q41, Q42, Q51, Q122]. The regional context confirms 'NOT FOUND' for translator regional/dialect background.

OC-4: Re-annotation by representative Ethiopian intercity travelers (urban Addis, regional Amhara, Amharic-as-L2 speakers) is needed; gap_id 5 confirms no existing L2-speaker NLP dataset to bootstrap from.

OC-5: Aggregation uses three judgment workers per utterance [Q63]; if the three-worker pool is demographically homogeneous (e.g., all urban Addis fluent speakers), L2-speaker and code-switched perspectives are systematically excluded. INSUFFICIENT DOCUMENTATION on judgment-worker demographics.

OC-6: Severe convergent and external-validity risks: labels do not correlate with regional stakeholder judgments for the linguistic phenomena (code-switching, L2 phrasing, calendar/operator slots) that the deployment must handle [Q6, Q104, Q107].

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Q2	“MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera.” (p.1)
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Q4	“MASSIVE was created by localizing the SLURP NLU dataset (created only in English) in a parallel manner.” (p.1)
Q6	“one difficulty in creating massively multilingual NLU models is the lack of labeled data for training and evaluation, particularly data that is realistic for the task and that is natural for each given language.” (p.1)
Q41	“The MASSIVE dataset was collected using a customized workflow powered by Amazon MTurk.” (p.5)
Q42	“We required a vendor pool with the capability and resources to collect a large multilingual dataset.” (p.5)
Q51	“We offered two mechanisms to vendors for evaluating workers to be selected for each language. The first, which was used to select workers for the translation task, was an Amazon MTurk-hosted fluency test where workers listen to questions and statements in the relevant language and were evaluated using a multiple-choice questionnaire.” (p.6)
Q63	“The output of the second workflow (the fully localized utterance) is judged by three workers for (1) whether the utterance matches the intent semantically, (2) whether the slots match their labels semantically, (3) grammaticality and naturalness, (4) spelling, and (5) language identification—English or mixed utterances are acceptable if that is natural for the language, but localizations without any tokens in the target language were not accepted.” (p.6)
Q104	“Second, there are some low-quality utterances, both in the seed data and in the translations.” (p.9)
Q107	“Relatedly, allowing the worker to decide on translation versus localization of slot entities added further noise to the dataset, although we try to store this decision in the metadata.” (p.9)
Q122	“By clicking "SUBMIT", I also certify that I am a native speaker or am fluent in the required language” (p.17)
WEB-1	Oromo speakers >40% of Ethiopia's population — large L2-Amharic share among intercity travelers (https://arxiv.org/html/2403.19365v1)
WEB-18	Amharic–Afaan Oromo bilingual code-switching corpus exists for hate speech only — domain mismatch with booking (https://link.springer.com/article/10.1186/s40537-024-01044-y)
WEB-20	Amharic/Oromo/Tigrigna code-switching empirically documented in Ethiopian social media — confirms the phenomenon is real and is unrepresented in monolingual MASSIVE Amharic (https://aclanthology.org/2024.rail-1.13.pdf)

Information Gaps

- Vendor identity and translator regional/dialect demographics for Amharic
- Distribution of judgment scores for Amharic utterances and how filtering by score affects size

Requires Expert Verification

- Empirical disagreement rate between MASSIVE Amharic labels and Ethiopian intercity-traveler relabels on a sample
- Whether code-switched Amharic–Oromo utterances would be rejected by MASSIVE's language-identification judgment [Q63]

Remediation

Gap 1: Amharic translator demographics undisclosed; no representation of L2-Amharic speakers or code-switching with Oromo/Tigrinya, despite user confirmation that these occur in real utterances

Recommendation: Recruit a regional annotator pool stratified across urban Addis, Amhara region, and L2-Amharic speakers from Oromia and Tigray corridors. Re-label a sample of MASSIVE Amharic utterances and measure inter-annotator agreement against original labels; then build a code-switched evaluation set drawing on the Amharic–Oromo corpus [WEB-18] and RAIL 2024 lexicon [WEB-20] adapted to booking-domain vocabulary.

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